ENG-0009 Incident Postmortem: INC-049

Impact

The incident involving the billing-gateway system (SYS-008) on April 16, 2025, resulted in a Sev2 severity classification. Given the mission-critical nature of the billing-gateway, the disruption potentially affected core billing operations and financial processing workflows. The incident lasted for approximately 10.5 hours, during which time service availability was compromised, impacting internal stakeholders and external clients reliant on timely billing data. Although the precise financial or client impact metrics are not available, the extended resolution time underscores the significance of the disruption and the need for process improvements to prevent recurrence.

Timeline

- **April 16, 2025:** The incident was reported by Stephanie Ross, an accountant within Project Cascade, indicating an operational anomaly in the billing-gateway system.
- **Initial Detection:** The incident was identified and escalated to the on-call engineer, Dylan Miller, a Procurement Specialist based in Singapore, who was responsible for immediate response.
- **Resolution Process:** Over the course of 10.5 hours, Dylan Miller investigated and addressed the root cause, which was classified under deployment_error.
- **Resolution:** The system was restored to normal operation by the end of the resolution period, with all services functioning as intended.

Root Cause

The root cause of the incident was identified as a deployment_error. This error originated during a recent deployment process within the billing-gateway system, which is built on a Python/FastAPI/PostgreSQL tech stack. The deployment mistake led to system instability, affecting the billing operations critical to Project Cascade, owned by department D-04.

Action Items

Action Item	Responsible Party	Due Date	Description
Conduct a comprehensive review of recent deployment procedures to identify gaps.	DevOps Team	Within 2 weeks	Ensure deployment protocols are robust and include validation steps to prevent similar errors.
Implement additional pre-deployment testing and automated checks for the billing-gateway system.	QA and Deployment Teams	Within 3 weeks	Reduce the likelihood of deployment errors through enhanced testing frameworks.
	Incident Response Team	Within 1 month	

Action Item	Responsible Party	Due Date	Description
Update incident detection and escalation procedures to enable faster response times.			Minimize downtime duration by improving early detection and escalation processes.
Schedule training for deployment best practices for all relevant engineering staff.	Engineering Management	Within 1 month	Ensure team members are equipped with the latest deployment protocols and error mitigation strategies.
Review and document the incident response timeline and lessons learned for continuous improvement.	Incident Postmortem Team	Within 2 weeks	Facilitate organizational learning and improve future incident handling.

Additional Notes

Dylan Miller, who was on call during the incident, played a critical role in the resolution process. The incident underscores the importance of rigorous deployment procedures, especially for mission-critical systems like billing-gateway within Project Cascade. Stephanie Ross, the reporter of the incident, provided timely notification that enabled prompt action. Moving forward, the focus will be on strengthening deployment controls and incident response protocols to mitigate the risk of similar incidents occurring again.